# Errors reveal the depth of human reasoning

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**Abstract**

How deeply people think before choosing is the hidden quantity at the centre of bounded rationality, yet rarely measured in natural behaviour. We read it from the structure of human mistakes: across more than a million skill-graded chess decisions scored by an engine at every search depth, the moves of strong and weak players look comparably good to a shallow analysis and separate only as it deepens—what divides them is the depth at which their errors are exposed, the depth of satisficing. It rises with skill in slow play, contracts as the clock tightens, and vanishes in the fastest games. Estimated without timing information, it tracks how long players thought, recovers the known depth of synthetic agents, and improves move prediction only where deep reasoning has room to matter. Every central result survived a prospective replication registered before its data existed. What people get wrong, more than what they get right, reveals how they think.

**Keywords:** bounded rationality; satisficing; depth of reasoning; decision-making; expertise; chess

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**Author note.** *Results report held-out analyses on the full dataset; strata and the primary interaction were fixed before held-out evaluation. Evaluation is split by player throughout. T. Maharaj formerly published as T. T. Biswas.*

# 1 Introduction

Seventy years of decision science rest on a quantity nobody can see. When Simon proposed that people *satisfice*—search until an option clears an acceptability threshold, then stop—he made the depth of that search the load-bearing variable of bounded rationality (Simon, 1955, 1956). The same hidden quantity does the work in the fast-and-frugal programme (Gigerenzer & Goldstein, 1996), in resource-rational analysis, which treats cognition as the optimal use of limited computation (Callaway et al., 2022; Griffiths, Lieder, & Goodman, 2015; Lieder & Griffiths, 2020; Russell & Wefald, 1991), in heuristics-and-biases accounts of judgement (Kahneman & Tversky, 1979; Tversky & Kahneman, 1974), and in the level-$k$ and cognitive-hierarchy models of strategic thinking (Camerer, Ho, & Chong, 2004; M. Costa-Gomes, Crawford, & Broseta, 2001; M. A. Costa-Gomes & Crawford, 2006; Crawford, Costa-Gomes, & Iriberri, 2013; Nagel, 1995; Stahl & Wilson, 1995; Strzalecki, 2014). Yet in practice, how far a person looks before committing has been inferred almost entirely from small staged experiments. A notable recent exception used a controlled board game with behavioural modelling to show that planning depth grows with expertise (van Opheusden et al., 2023). Whether such regularities hold in real, repeated, consequential choice—outside the laboratory and at the scale of populations—has been hard even to ask, because natural decisions rarely come with a ground truth for how good each option was, let alone for how good it *looked* at each successive depth of consideration.

Chess supplies exactly that, which is why it has served for a century as the model organism of expert cognition (Chase & Simon, 1973; de Groot, 1965; Ericsson, Krampe, & Tesch-Römer, 1993; Gobet & Simon, 1996; Newell & Simon, 1972). Millions of decisions by players of precisely known skill are recorded in real competition, and every one of them can be scored by an engine far stronger than any human (Campbell, Hoane Jr., & Hsu, 2002; Silver et al., 2018)—not once, but at every depth of an iterative-deepening search. The engine's opinion of a move changes as it looks deeper: a move that seems strong at a glance may be refuted only far down the tree, and a move that looks dull at first may reveal its point three moves later. This depth-resolved record lets us put a new question to every human mistake: *at what depth does the flaw become visible?* And therefore: how deep must the player have looked?

Whether expertise involves searching deeper at all is itself an old and unsettled question in this model system. De Groot's think-aloud protocols found that grandmasters and strong club players searched to similar depths and considered similar numbers of moves; what set the masters apart was that they looked at the right moves (de Groot, 1965). Later studies with more players and finer measures found that depth does grow with skill (Campitelli & Gobet, 2004; Charness, 1981; Holding, 1985), though modestly and unevenly (Gobet, 1998), and a re-examination seventy years on concluded that search and pattern knowledge are complementary rather than rival explanations (Connors, Burns, & Campitelli, 2011). Every one of these studies rests on a few dozen players thinking aloud over a handful of positions. Engines have since been used to score human moves against a near-perfect standard (Anderson, Kleinberg, & Mullainathan, 2016; Guid & Bratko, 2006; Regan & Haworth, 2011), but the depth at which a human's error becomes visible—the quantity the classic debate turns on—has not been read from that standard at scale.

We call the answer the **depth of satisficing**—the depth at which the move a player chose stops looking better than the move they should have played. Our central empirical claim is that this is the dimension along which skill is most legible. Judged only at shallow depth, the moves of players separated by a thousand rating points look comparably good; the gap opens as the engine searches deeper. What divides them is carried by their errors—not by how many they make, but by how deep one must look before the flaw appears. Skill is present even in moves played at a glance, since stronger players recognise better options without searching; but most of what rating buys is realised only through the depth to which a player effectively searches.

Online chess also supplies a natural, graded manipulation of how deeply a person can afford to look: the time control. A player with an hour for the game can follow a line many moves ahead; one with ten minutes cannot; and one playing *bullet*—a minute or two for all the moves—has time only to react. Because the same players compete across every control, the clock varies the budget for search while holding the person, and their pattern knowledge, fixed. If the depth of satisficing measures search rather than mere skill, it should track that budget: deepest in classical play, shallower in rapid and blitz, and collapsing in bullet, where even strong players must move on recognition alone with no time to verify what they see. This graded prediction across four controls is among the sharpest tests we can put to the construct, and its limiting case is a built-in control: in bullet, search is unavailable to everyone, so any residual skill difference there is pattern, not depth.

Three things distinguish this work from earlier studies of chess errors and engine depth (Biswas & Regan, 2015; McIlroy-Young, Sen, Kleinberg, & Anderson, 2020; Regan & Haworth, 2011). First, we replace descriptive curve-fitting with a generative model: the player is a latent distribution over search depths, so the depth of satisficing is a posterior quantity with stated uncertainty rather than the crossing point of two fitted curves. Second, we do not assume that engine depth is a valid proxy for human reasoning depth; we test it—predictively, convergently, and differentially. Third, we subjected the construct to an adversarial replication programme: hypotheses, pipeline, and decision rules were pre-registered and then confirmed on untouched months of data, ending with a fully prospective test registered before its data existed (Nosek, Ebersole, DeHaven, & Mellor, 2018). The final, registered confirmation succeeded on all three primary hypotheses.

## 2 Results

**Reading a decision at increasing depth.** A chess engine evaluates by *iterative deepening*: it scores every candidate move looking one ply (half-move) ahead, then two, and so on, revising each move's evaluation at every new horizon; evaluations change with depth because deeper search uncovers consequences that shallower search cannot see—the game-tree analogue of thinking further ahead. A move's value is therefore not a number but a *trajectory* across depth, and that trajectory is our raw material. Figure 1a shows a real decision from the data. A 1910-rated player, offered a bishop, recaptured it (fxe6): at shallow depth the capture looks close to winning, but from about eight plies the refutation comes into view and its value collapses. The quiet king move Kb8 shows the opposite signature—unimpressive at a glance, it proves best once the search runs deep. Writing $\delta_{m,d}$ for move $m$'s *regret* at depth $d$—the win probability it concedes relative to the best option visible at that depth (Methods)—we call the signed area between a move's regret trajectory and its full-depth endpoint the move's *swing*, $\mathrm{sw}(m) = \sum_d(\delta_{m,d} - \delta_{m,D})$: positive (*swing-up*, a deep discovery) when a move looks worse shallow than it truly is, negative (*swing-down*, a trap) when shallow search flatters it. A position's total swing, $\sum_m|\mathrm{sw}(m)|$, measures how much the picture changes with depth—how much deliberation the position can reward. The depth at which the played move's flaw or worth becomes visible is the signal everything below is built on.

**Skill lives in the errors, not the successes.** Our primary sample comprises 922,412 decisions by 16,328 players rated 1200 to 2600+, drawn from online play and from elite over-the-board tournaments (Methods). At shallow engine depth, the average error of the played move barely distinguishes a club player from a grandmaster; the gap opens only as the engine looks deeper (Fig. 1b). Because online ratings are pool-specific—a separate rating per time control—we read every skill comparison *within* a control: at full depth, mean regret of the played move falls with rating in classical (0.059 at the weakest band to 0.028 at the

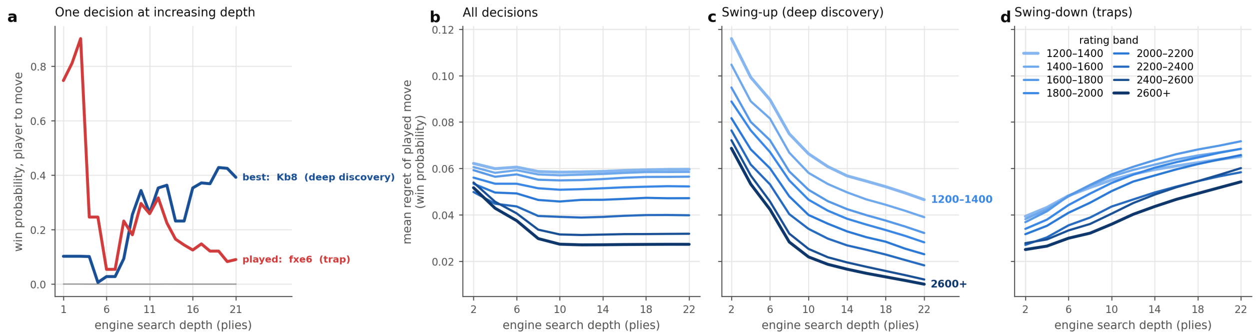


Figure 1: **Skill separates only at depth, and mostly on deep-discovery moves. a**, A worked example: candidate-move win probabilities across engine search depth for one real decision (grey, other candidates; moves in standard algebraic notation). The played pawn capture fxe6 looks nearly winning at shallow depth and collapses once the refutation surfaces (a *trap*, swing-down); the quiet king move Kb8 reveals its worth only deep (a *deep discovery*, swing-up). **b–d**, Mean regret of the played move across engine search depth, by rating band (light to dark blue: 1200–1400 up to 2600+), within the slow controls (classical and rapid). **b**, All decisions: rating bands are nearly indistinguishable at shallow depth and fan apart as the engine looks deeper. **c**, Swing-up (deep-discovery) decisions: a clean, well-ordered rating fan—the rating signal concentrates here. **d**, Swing-down (trap) decisions: bands bunch and overlap, separating roughly half as much. The rating axis spans online play (lower bands) to elite over-the-board play (upper bands).

strongest), in rapid, and in blitz, but stays nearly flat in bullet (0.070 to 0.055), where no one has time to search deeply. The separating information, moreover, is not spread evenly across mistakes. It is concentrated in *deep-discovery* decisions—swing-up moves whose worth only emerges at depth—which fan out cleanly by rating (Fig. 1c), while trap-like swing-down decisions bunch the rating bands together (Fig. 1d). A single swing-up decision carries roughly twice the rating information of a swing-down one (pool-normalised single-decision $R^2 \approx 0.006$ versus 0.003), an asymmetry that recurs in every replication month below. What matters is not how large an error is, but at what depth the move's true worth appears.

**Depth is not the size of the error.** That claim can be tested directly, without the model. For every erroneous move (full-depth regret $\geq 0.05$) define its *exposure depth* as the first search depth at which its regret reaches half its final value—the point at which the flaw comes into view (Methods). Exposure depth is essentially uncorrelated with how large the error eventually proves to be (Spearman $\rho = +0.02$ in the primary month, and between

+0.01 and +0.03 in every replication month): how far one must look to see a mistake and how much it costs are separate dimensions. And it is exposure depth that carries the skill signal. Holding error magnitude fixed, stronger players' errors are exposed deeper. Among the largest blunders (regret $> 0.20$), exposure depth rises by 0.93 plies per 1,000 rating points (player-clustered SE 0.07), from about 3.8 plies below 1600 to 4.8 above 2000, and the slope replicates at 1.07, 0.85 and 0.78 plies in the three replication months (all $z > 5$). The effect is graded by severity—weak for small inaccuracies, strongest for large blunders—so the errors that distinguish a strong player from a weak one are not smaller but *better hidden*: mistakes a shallow search would not catch. Skill is legible in where an error surfaces, not in its size.

**Depth of satisficing rises with skill and collapses under time pressure.** Fitting the latent-depth model (Methods) and reading off each decision's posterior expected depth, the construct behaves exactly as a measure of deliberation should—provided, again, that one looks within a time control, since pooling would conflate skill with available clock. Within classical games, the depth of satisficing climbs from about 10.8 plies in the lowest rating band to about 13.7 in the highest, with non-overlapping player-clustered 95% intervals at the extremes (Spearman $\rho = +0.40$; rapid +0.66; blitz +0.45; Fig. 2a). Engine plies are an ordinal ruler for effective depth, not literal look-ahead: a Stockfish search to depth 12 is heavily pruned and extended, and think-aloud studies put average human search at only a few plies (Charness, 1981; de Groot, 1965). What the scale supports is comparison—who satisfices deeper than whom, and when—and that is how it is used throughout. Bullet is the built-in negative control: the relationship vanishes ($\rho = -0.04$, flat across bands). When nobody can search, depth stops separating anybody, which rules out the construct being a re-description of rating. The same signature appears *within* players: de-meaned depth rises across think-time quartiles, from $-0.14$ plies on a player's fastest quarter of moves to $+0.33$ on their slowest (Fig. 2b).

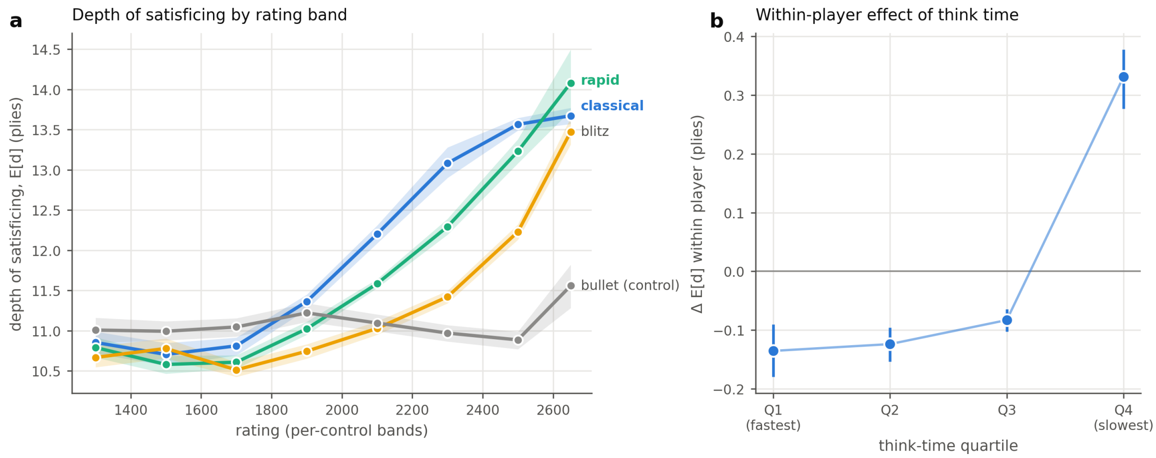


Figure 2: **Depth of satisficing rises with skill and with time to think.** **a**, Posterior depth of satisficing by rating band, within each time control (shaded bands, player-clustered bootstrap 95% CIs). Depth rises with rating in classical (Spearman $\rho = +0.40$), rapid (+0.66) and blitz (+0.45); bullet, where no one can search, is the flat negative control ($\rho = -0.04$). **b**, Within-player effect: de-meaned depth by think-time quartile (dots, pooled means; bars, player-clustered bootstrap 95% CIs). The same players search deeper on their slower moves.

**What the clock buys.** The four controls form a graded ladder of thinking time rather than arbitrary labels. In the online sample the median time a player actually spends on a move falls from about 11 seconds in classical play to 5 in rapid, 3 in blitz, and 1 in bullet—close to an order of magnitude, tracking base budgets from thirty minutes down to one. This ladder is what the depth of satisficing responds to: with more clock a player can carry a line further before committing, so depth rises with skill in classical, rapid, and blitz, where there is time to search, and goes flat in bullet, where a move must be produced in about a second—from recognition rather than search. The within-player pattern says the same thing from the other side: the same person searches deeper on their slower moves (Fig. 2b). The clock sets the ceiling on attainable depth, and skill is expressed in how fully a player uses the room the clock allows—which is why time pressure compresses the skill gap and slow play opens it.

**Inferred depth tracks real thinking time it was never shown.** The construct's key non-circular test removes the clock from the model entirely. Refit with the clock feature

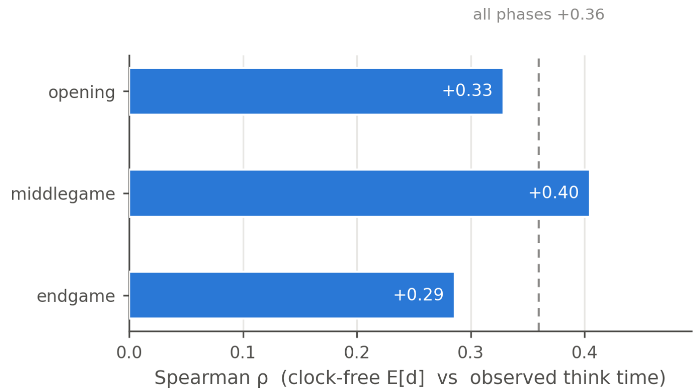


Figure 3: **Depth inferred without the clock tracks real thinking time.** Held-out Spearman correlation between the clock-free model's per-decision depth and observed time-on-move, by game phase (dashed line, all phases pooled). The correlation peaks in the middlegame, where deliberation varies most.

zeroed, so that timing can play no role in the estimate, the model still assigns held-out depths that correlate with how long the player actually spent on the move (Spearman $\rho = +0.36$ overall), and the correlation peaks in the middlegame (+0.40)—the phase where deliberation genuinely varies most—falling to +0.33 in the opening and +0.29 in the endgame (Fig. 3). Inferred depth recovers observed deliberation without ever being told about time. That is convergent evidence that the posterior reflects thinking, not the rating prior it partly leans on.

**A depth-aware model predicts human moves better, exactly where depth should matter.** If the depth of satisficing is real, knowing it should help predict what a human will play, and the help should be confined to situations where depth can vary and can matter. Both hold. A fusion of a state-only human-move model (Maia-3 (Monroe, Eilender, Chalmers, Tang, & Anderson, 2026)) with a mixture-over-depths search model improves held-out cross-entropy over the state-only model by a modest 0.012 nats overall (95% CI $[0.008, 0.016]$, player-clustered bootstrap; top-1 accuracy +0.7 points). The scientific content is not the average but the pre-registered differential: the gain concentrates in classical, high-swing positions (+0.053 nats) and is absent or negative in fast or low-swing play (Fig. 4). The swing-magnitude $\times$ time-control interaction is +0.060 nats (95% CI $[0.056, 0.064]$),

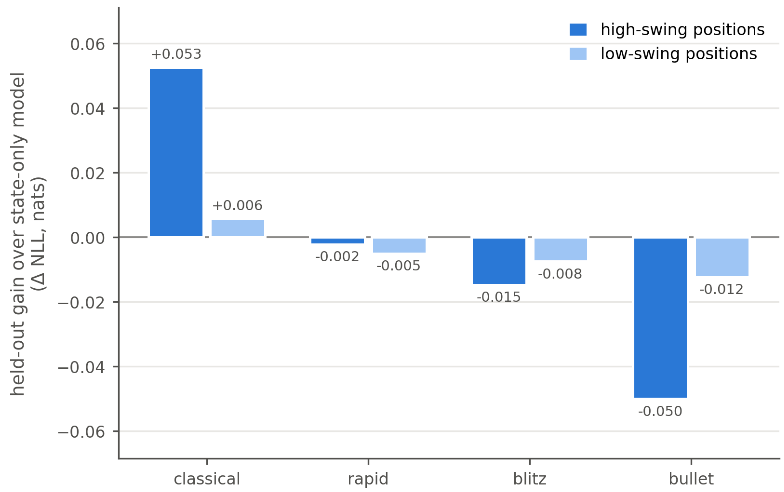


Figure 4: **Depth improves move prediction only where depth can matter.** Held-out gain of the depth-aware fusion over the state-only model (nats), by time control and position swing. The gain concentrates in classical, high-swing positions and is absent or negative in fast or low-swing play—the pre-registered differential.

confirmed by a mixed-effects model with players as random effects ($\beta = 0.058$, $p < 10^{-277}$). No state-only model can produce this pattern, however strong (Ruoss et al., 2024); within classical play the gain falls almost entirely on swing-up moves (+0.048 nats)—precisely the choices whose worth a pattern model cannot see because it does not look ahead.

**Positions are test items; swing is their discrimination.** These findings admit a measurement-theoretic reading that unifies them. Treat each position as a test item whose *difficulty* is its critical depth—the deepest ply at which a misleading move still looks best—and whose graded *response* is the severity of the move played (best, inaccuracy, mistake, blunder). An explanatory graded-response model (Samejima, 1969), with item properties entered as engine-derived covariates in the manner of a linear logistic test model (Fischer, 1973) (each natural position occurs only once, so per-item parameters are unidentifiable), recovers the full expected structure: error severity falls with ability ($\beta = -0.13$, $p < 10^{-80}$), rises with item difficulty ($\beta = +0.06$, $p < 10^{-24}$), and—the decisive term—ability discriminates more sharply in high-swing positions (ability $\times$ swing $\beta = -0.08$, $p < 10^{-38}$; Fig. 5a). In item-response language, *swing is item discrimination*: test information concentrates in

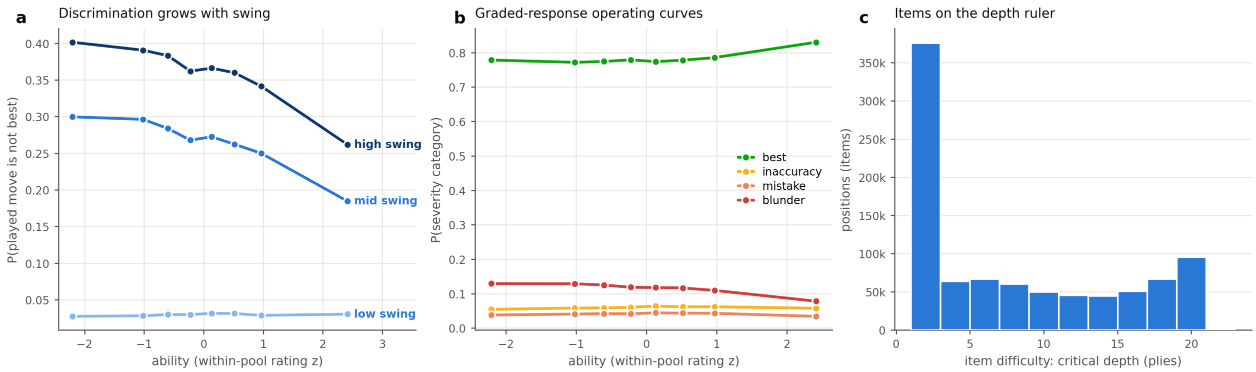


Figure 5: **An item-response reading: swing is item discrimination.** **a**, Probability that the played move is not the engine-best, versus ability (within-pool rating $z$), by position-swing tercile: ability discriminates far more steeply in high-swing positions. **b**, Graded-response operating curves: probability of each severity category versus ability; P(best) rises and P(blunder) falls monotonically. **c**, Items placed on the ply ruler by their critical depth—the same scale on which the depth of satisficing measures person ability.

high-swing decisions, which is exactly why skill is legible in so small a fraction of moves. The graded operating curves shift cleanly along the ability axis (Fig. 5b), and item difficulty lives on the same ply ruler as the person-side depth of satisficing (Fig. 5c)—a natural generalisation of rating systems, which are themselves one-parameter item-response models on game outcomes (Elo, 1978; Embretson & Reise, 2000; Rasch, 1960). The structure is not a platform artefact. On an independent pool of 99,358 chess.com decisions by 945 players—a different site, a different rating system—the same model reproduces every sign (ability $\beta = -0.08$, difficulty $+0.11$, swing $+0.87$, ability $\times$ swing $-0.06$; all $p < 10^{-13}$), and the same holds on a second Lichess month (2026-04: ability $\times$ swing $\beta = -0.09$, $p < 10^{-40}$)—three independent pools in all. At the present decision density per player, ability is reliable only in aggregate; person-dense, pool-invariant ability estimation is future work.

**Profiles, not a single number, separate expert styles.** Averaged per player from an *elo-free* refit (so that recovering rating is not circular), four behavioural axes—depth of satisficing, trap susceptibility, deep-discovery rate, and time elasticity—summarise a player's style. Across 873 players with at least 100 decisions, the four-axis profile recovers within-pool rating at $R^2 = 0.10$ under nested cross-validation while the depth axis alone recovers none

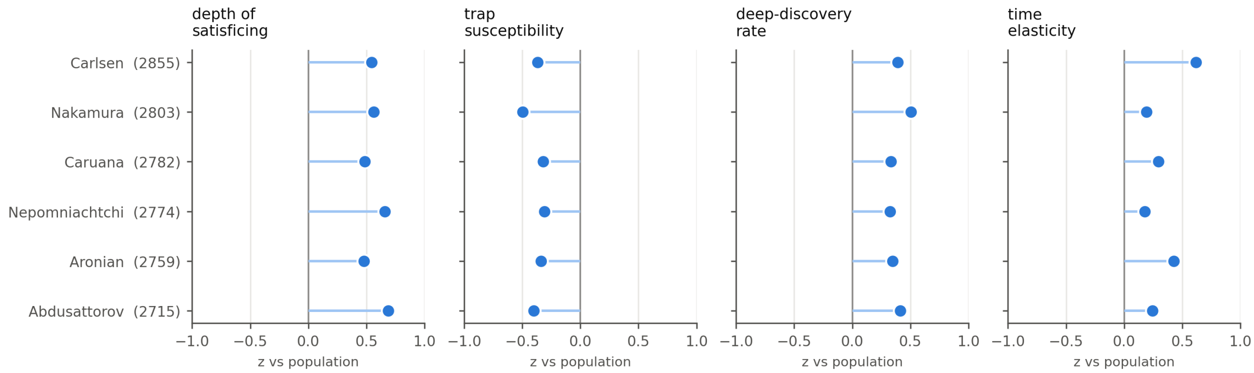


Figure 6: **Four behavioural axes separate elite styles that one number confuses.** Profile z-scores (versus the full population) for the six elite players with the most analysed decisions, ordered by rating. A one-dimensional depth score inverts elite ratings (9 of 15 pairs discordant); the four-axis profile recovers within-pool rating across 873 players ($R^2 = 0.10$, nested cross-validation) where depth alone recovers none ($R^2 \approx 0$), and reads the styles directly—for example Nakamura as most trap-resistant but among the least time-elastic, Carlsen as converting think time into depth most steeply.

($R^2 \approx 0$): the added axes carry the signal, reported on the honest within-pool scale rather than the cross-pool one (which flatters recovery by conflating skill with pool membership). Among elite players the profile also resolves a known embarrassment of one-dimensional scores: a bare depth ranking inverts the true order of the world's best (9 of 15 elite pairs discordant), ranking, for example, Abdusattorov and Nepomniachtchi above Carlsen. The four axes instead attribute each player's strength differently—Nakamura is the most trap-resistant of the six yet among the least time-elastic (a blitz specialist who reaches his depth *fast*), while Carlsen converts thinking time into depth more steeply than anyone—and the inversions disappear (Fig. 6).

**The instrument recovers depth it has never seen.** Synthetic agents that satisfice at fixed, known depths, played through the identical pipeline, come back in the right order: recovered depth increases monotonically with planted depth (group-level Spearman = 1.0 across planted depths 4–16), and the estimator independently recovers the agents' decision sharpness ($\hat{\beta} = 7$ against a generating $\beta = 6$; Fig. 7). Recovery is ordinal rather than metric—per-agent estimates are noisy and compressed toward the middle (mean absolute

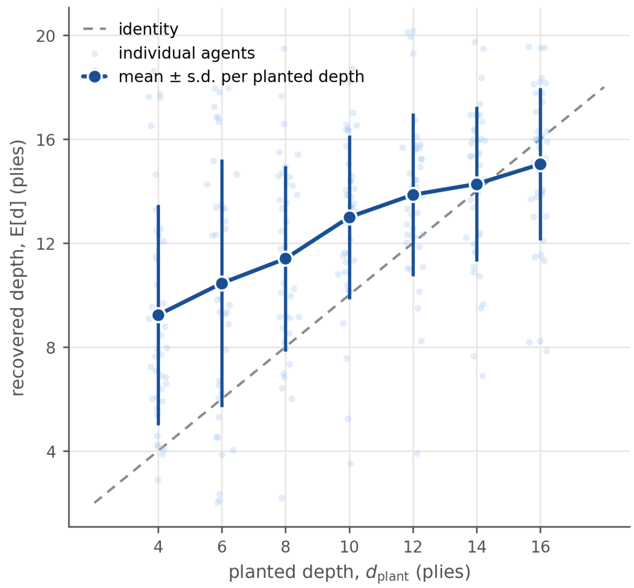


Figure 7: **The instrument recovers planted search depth.** Synthetic agents satisficing at fixed depths (4–16 plies), run through the identical pipeline. Pale dots, individual agents; dark points, mean ± s.d. per planted depth; dashed line, identity. Recovery is monotone (group-level Spearman = 1.0) though ordinal: per-agent estimates are compressed toward the middle (MAE 3.7 plies at ∼70 decisions per agent).

error 3.7 plies on roughly 70 decisions per agent)—but it establishes that the model measures depth rather than position difficulty. Recovery requires a flat prior over depth: imposing the human rating-conditioned prior collapses the estimate toward its mean, which is why the two non-circular tests above, and not the posterior's absolute scale, carry the evidential weight.

**A pre-registered programme ending in a prospective confirmation.** We pre-registered hypotheses, pipeline, and decision rules, and then ran the frozen pipeline once on each of three untouched Lichess months (Nosek et al., 2018). The first replication (2026-05; 179,360 decisions; OSF KB4ZQ) confirmed the non-circular core—clock-free depth tracked thinking time ($\rho = +0.30$, middlegame peak), and the registered swing × time-control interaction held (+0.049 nats, mixed-effects $p < 10^{-36}$)—but failed its own strict rule because one arm of one primary (blitz, registered as positive) came out flat, and it exposed a swing-label convention error: the single-decision rating signal sits in swing-*up* decisions, not swing-down as first registered. We disclosed both, corrected the hypotheses, and registered a second, fully prospective protocol (OSF A6NYK) whose primary dataset—Lichess 2026-06—did not yet exist at registration. A robustness month processed under the same registration (2026-04; 168,654 decisions) meanwhile reproduced the slow-control rating relationship (classical $\rho = +0.47$, rapid $+0.54$), the convergent-time result ($+0.40$), and the interaction ($+0.076$ nats, $p < 10^{-48}$).

The prospective month then arrived and was processed once. All three primary hypotheses replicated, satisfying the registered decision rule. On 203,248 decisions from 3,707 players: depth rose with rating within every slow control (classical $\rho = +0.23$, 95% CI $[+0.15, +0.35]$; rapid $+0.72$ $[+0.59, +0.79]$) with bullet null as predicted ($+0.08$ $[-0.03, +0.19]$); clock-free depth tracked observed thinking time ($\rho = +0.32$, middlegame peak $+0.36$); and the depth-aware model beat the state-only model overall ($+0.010$ nats, 95% CI $[+0.003, +0.017]$) with the registered interaction intact ($+0.068$ nats $[+0.055, +0.079]$; mixed-effects $\beta = +0.059$, $p < 10^{-42}$). Two secondary outcomes are worth reporting straight. Blitz, which we had over-corrected to "null" after its flat 2026-05 showing, came back clearly positive ($\rho = +0.39$ $[+0.25, +0.50]$); across the four months the blitz relationship now reads $+0.45$, $\approx 0$, $+0.19$, $+0.39$—real but unstable, in contrast to the robust slow controls and the reliably null bullet. And the deep-discovery asymmetry replicated in its corrected direction (swing-up $R^2 = 0.0049$ versus swing-down 0.0027), as it has in every month examined. The construct's central claims thus rest on a confirmation that was, in the strict sense, predicted in public before the behaviour occurred.

# 3 Discussion

Errors are usually the noise of behavioural data. Here they are the signal. Where success is widely shared, the fine structure of failure—the depth at which a flaw becomes visible—turns out to be a stable, measurable signature of how a person reasons: it scales with expertise, contracts under time pressure, tracks deliberation it was never shown, and improves prediction exactly in the regime where deep reasoning has room to operate. The pattern supports a specific mechanism: people commit to the best option found at the search horizon they reach, rather than integrating over the whole trajectory—consistent with theoretical results that level-$k$ predictions are independent of higher-order belief tails (Strzalecki, 2014), and connecting satisficing to model-based control (Daw, Niv, & Dayan, 2005), decision-tree pruning (Huys et al., 2012), and bounded evidence accumulation (Ratcliff & McKoon, 2008). Where planning depth has been measured before, it required controlled laboratory games and explicit search models (Callaway et al., 2022; van Opheusden et al., 2023); here the same construct is read non-invasively from natural, high-stakes play at population scale, against a depth-resolved value standard, and it survives a prospective registered test. The item-response formulation adds a measurement-theoretic dividend: positions become items with engine-derived difficulty and discrimination, so people and problems share one ply-denominated scale—rating systems generalised from game outcomes to individual decisions.

The result also speaks to the oldest question in the psychology of chess: whether experts search deeper or merely search better (Charness, 1981; Connors et al., 2011; de Groot, 1965; Gobet, 1998). The depth of satisficing measures *effective* depth—how far the chosen move survives scrutiny—and is deliberately agnostic about whether that depth is reached by looking further down a line or by selecting, from pattern, lines that do not fail early. The data suggest both operate and can be told apart. Skill is visible even on moves played in a second, where only pattern can act; yet expert errors of a given cost are better hidden rather than smaller, and, in a supplementary analysis, the skill gradient in move quality roughly doubles when players deliberate rather than move at once. De Groot's null and Charness's positive finding are then reconciled by the clock: with little time, strong and weak players look alike in depth and differ in pattern; given time, the strong convert it into depth and the weak cannot.

The individual profiles suggest what the construct is for. Expertise is not a scalar: two players of equal rating can owe their strength to depth, to trap resistance, to deep discovery, or to the efficiency with which time becomes depth—echoing evidence that expert flexibility itself varies (Bilalić, McLeod, & Gobet, 2008). An agent that estimates a person's depth of satisficing could, in principle, select problems pitched just beyond it; a rating system built on decisions rather than outcomes could equate skill across pools that never play each other.

The limitations are the flip side of the design. Engine search depth is a *validated proxy* for human selective search—validated by convergence with thinking time and by recovery of planted depth—but it is not isomorphic to human search, which is narrow, uneven, and pattern-guided. The depth-of-satisficing posterior leans partly on a rating-conditioned prior; per-decision move evidence is weak, which is why the clock-free and synthetic tests, not the posterior's absolute scale, carry the claims. The average predictive gain is small, and the case rests on its registered differential rather than its size. The blitz rating–depth relationship is genuinely variable across months and should not be quoted as a stable effect. And chess players are a highly trained, self-selected population in an adversarial task; the approach extends in principle to any domain with skill-graded decisions and a depth-resolved value standard—a value network queried at increasing depth in Go, or staged value estimates in planning tasks—but that generality remains to be demonstrated.

# 4 Methods

**Data.** The primary dataset comprises 922,412 decisions by 16,328 players: online games from the Lichess open database (month 2025-09), balanced-sampled across rating bands (200-point bins, 1200–2600+) and time classes (bullet, blitz, rapid, classical), with per-move clock times retained. Time classes follow Lichess's convention, by estimated game duration (initial time plus 40× the increment): bullet under 3 minutes, blitz 3–8, rapid 8–25, and classical 25 minutes or more. Each class therefore pools nearby formats—1+0 and 1+1 bullet, for instance, both fall in the bullet class despite an appreciable difference in effective time—so residual within-class heterogeneity remains, and every skill comparison in this paper is made within class, never across; plus an elite over-the-board stratum of 371,279 decisions from 1,261 titled players in broadcast tournament games, which supports the individual-profile case study. Replication datasets—Lichess 2026-05 (179,360 decisions), 2026-04 (168,654), and the prospective 2026-06 (203,248 decisions, 3,707 players, 113,421 classical)—were each sampled by the same frozen pipeline (identical seed, cell targets, and engine settings) and processed once. The independent-pool item-response check uses 99,358 decisions by 945 titled chess.com players harvested via the public Published-Data API. Synthetic agents (below) provide ground truth. All decisions are restricted to plies 9–120 to skip opening-book play; no further selection is applied, so every non-book decision in a sampled game is analysed and the sample is not conditioned on position difficulty or on the quality of the move played.

**Engine analysis and value scaling.** Positions are analysed with Stockfish 18 in multi-PV mode, recording each candidate move $m_i$'s evaluation at every depth $d = 1 \ldots D$ ($D$=21, $K$=9). The candidate set is the union of the top-$K$ across all depths, so trap moves keep a full trajectory. Because consecutive plies of each game are analysed, the value of a played move falling outside the top-$K$ is read off from the position it leads to (the next analysed decision): its win probability at depth $d$ equals one minus the successor's best win probability at depth $d-1$, matching the other candidates' negamax depth. Where no analysed successor exists (game end, final sampled ply) the played move takes the weakest retained candidate's win probability at each depth. With `UCI_ShowWDL` the engine reports win/draw/loss counts $(w, \mathrm{dr}, \ell)$ per mille, giving the win probability $P_{i,d} = (w + \frac{1}{2}\,\mathrm{dr})/1000$ directly—no logistic or centipawn scaling. Depth-$d$ regret is

$$\delta_{i,d} = \max_j P_{j,d} - P_{i,d} \;\geq\; 0. \tag{1}$$

Being a difference against the best move *at each depth*, $\delta$ is invariant to anything that shifts all moves together (position difficulty, evaluation scale), isolating the choice problem. Searches run to depth $D$ or a $2 \times 10^8$-node cap, whichever comes first; the 0.09% of positions that stop short have their unreached depths masked in the likelihood (reached depth recorded) rather than imputed.

**Pattern prior.** The human-pattern prior $q_i(p, c)$ is the Maia-3 policy (Monroe et al., 2026) at the side-to-move's rating, read from the policy head as a full distribution over legal moves.

**Latent-depth choice model.** A player who has effectively searched to depth $d$ chooses by a soft-max over depth-$d$ regret fused with the pattern prior as a product of experts,

$$P(\text{move}{=}i \mid d, p, c) = \frac{\exp(-\beta\, \delta_{i,d} + \alpha\, \log q_i)}{\sum_j \exp(-\beta\, \delta_{j,d} + \alpha\, \log q_j)}. \tag{2}$$

Effective depth is latent with distribution $\pi_d(p, c)$, emitted by a small network over a coarse depth grid conditioned on context $c$ (rating, time class, clock, move number, total swing). The move likelihood marginalises depth, $P(\text{move}{=}i \mid p, c) = \sum_d \pi_d(p, c)\, P(\text{move}{=}i \mid d, p, c)$, and parameters minimise the negative log-likelihood of the played move with an entropy penalty on $\pi$ for identifiability and $\beta$ tied globally. Data are split by player throughout.

**Depth of satisficing.** The posterior over effective depth given the observed move,

$$r_d = P(d \mid y, p, c) = \frac{\pi_d\, P(y \mid d)}{\sum_{d'} \pi_{d'}\, P(y \mid d')}, \qquad \hat{d} = \sum_d d\, r_d, \tag{3}$$

gives a per-decision depth with uncertainty. Lichess ratings are pool-specific (a separate Glicko-2 rating per time control, not comparable across controls), so no analysis pools ratings across controls; doing so would also confound skill with available clock. Per-band estimates carry player-clustered bootstrap 95% intervals (1,000 resamples of players) (Efron, 1979).

**Exposure depth.** For the model-free discriminant test, an erroneous move (full-depth regret $\geq 0.05$) has exposure depth equal to the first grid depth at which its regret reaches half of its full-depth value. Within bins of full-depth regret (0.05–0.10, 0.10–0.20, >0.20), exposure depth is regressed on rating (per 1,000 points) and move number with player-clustered standard errors, on online classical and rapid decisions; the same script is run unchanged on each replication month.

**Relation to the curve-intersection estimator.** The construct originates in a population-level heuristic (Biswas & Regan, 2015): average the played move's error and the engine move's error against depth, and read the depth of satisficing off the curves' crossing. That estimator returns one depth per cohort, carries no per-decision uncertainty, presupposes a single time control, and inherits the fit (cohort depths move by roughly a ply under reasonable fitting choices). We keep the underlying signal—a played move whose error grows with depth is exactly the move *swing* of Biswas and Regan (Biswas & Regan, 2015)—and replace the crossing with the generative posterior $r_d$: a per-decision quantity with credible intervals that supports within-control and within-player contrasts and is identifiable against synthetic ground truth. The population curve is stable only because it averages thousands of weak per-move signals, which the crossing view conceals and our identifiability test makes explicit: recovered depth is ordinal and prior-compressed, which is why the convergent and identifiability tests, not the posterior's absolute scale, carry the claims. Where the two views can be compared they agree—stronger players satisfice deeper and err less—now within time control, at far larger scale, on win-probability evaluation, with a human-pattern prior, and under a registered replication programme.

**Item-response formulation.** Each decision is an item. Difficulty $b_j$ is the deepest grid depth at which the apparent-best move differs from the full-depth best move—the critical depth $d_i$ of Biswas and Regan (Biswas & Regan, 2015), in plies; the ordered response is the played move's full-depth regret binned as best ($\leq 0.02$), inaccuracy ($\leq 0.05$), mistake ($\leq 0.10$), or blunder ($> 0.10$) in win-probability units. We fit an explanatory proportional-odds graded-response model (Fischer, 1973; Samejima, 1969) with ability (within-pool rating

$z$), $b_j$, position swing, and ability $\times$ swing as predictors; item properties enter as covariates rather than free parameters because most natural positions occur exactly once. The MLE uses a random subsample; response curves are computed on all decisions. Per-player ability has low split-half reliability at this decision density, so effects are reported at the item and rating-band level.

**Profiles and rating recovery.** Per player with at least 100 decisions: mean $\hat{d}$; trap susceptibility (rate of played swing-down moves, weighted by shallow attractiveness); deep-discovery rate (rate of played swing-up moves); and time elasticity (within-player slope of $\hat{d}$ on log think-time). The depth axis comes from a model fit with the rating feature removed, keeping recovery non-circular. Rating is predicted from the standardised profile by ridge regression under nested five-fold cross-validation (inner folds select the penalty), within pool (z-scored per time control, over-the-board play its own pool). The elite case study reports the standardised axes for the six broadcast players with the most analysed decisions.

**Model comparison and the registered differential.** State-only ($\beta$=0), search-only ($\alpha$=0), and fusion predictors are fit under one player split and compared on held-out cross-entropy and top-1 match. The registered hypothesis is an interaction: the fusion's gain over state-only is larger in high-swing than low-swing positions and in classical than blitz. It is tested with a player-clustered bootstrap and a linear mixed-effects model with player random intercepts.

**Convergent and recovery validation.** Convergent: refit with the clock feature zeroed, then correlate held-out $\hat{d}$ with observed time-on-move, by game phase. Recovery: agents play a soft-max ($\beta_{\text{gen}}$=6) over fixed-depth win probabilities; their games pass through the identical pipeline, and a per-agent flat-prior posterior over depth (search likelihood, with the population $\beta$ fit) is compared with the planted depth.

**Pre-registered replication programme.** Hypotheses, pipeline, sampling seed, and the decision rule (success iff all three primary hypotheses replicate directionally with stated intervals) were registered before analysing each replication month (first registration: OSF KB4ZQ; corrected and prospective second registration: OSF A6NYK, whose primary month 2026-06 was registered before Lichess published the data). Each month was processed exactly once by the frozen pipeline. Deviations are disclosed in Results: the first registration's blitz prediction was too strong and its swing-direction convention reversed; the second registration's blitz-null prediction proved too weak in turn (blitz positive in the prospective month). Neither deviation touches the primary hypotheses.

**Reproducibility.** Stockfish 18 (shared NNUE), Threads 1 per process across many single-thread engines, Hash 256 MB, $D$=21, $K$=9, $2\times10^8$-node cap, driven through the `python-chess` UCI bridge (v1.11), which also provides all board and move representation; Maia-3 (`maia3-79m`) (Monroe et al., 2026); depth grid $\{2, 4, \ldots, 22\}$; sampling seed 17 and 60 games per (rating band $\times$ time class) cell in every month; train/test split by player; model fits seeded. All analyses ran on a single dual-socket 64-core server with one GPU; the engine stage dominates the compute at roughly 1,400 core-hours per month of data, and all model fitting and analysis complete within about an hour. Strata and the primary interaction were fixed before held-out evaluation; post-hoc choices are labelled exploratory.

**Ethics.** This study is a secondary analysis of publicly available, pseudonymous game records; no participants were recruited, no interaction or intervention took place, and no private or identifiable personal information was collected or processed. Online players appear only under their self-chosen public usernames; the named elite players analysed in the individual profiles are public figures whose over-the-board tournament games are published as a matter of course. As research limited to secondary analysis of public data, the study did not require prior ethics approval and no formal ethics review was sought; informed consent was not required for the same reason.

**Data availability.** All game records derive from public sources: the Lichess open database (https://database.lichess.org; months 2025-09, 2026-04, 2026-05, 2026-06), openly licensed Lichess broadcast exports of over-the-board play, and public games retrieved via the chess.com Published-Data API (https://api.chess.com). The derived analysis datasets—per-position engine depth trajectories, Maia-3 policies, and the model tensors behind every result, for the primary month and all three replication months—are deposited in the study's Open Science Framework project (https://osf.io/ruhy8) under a CC-BY licence; the deposit DOI will be minted on publication.

**Code availability.** The full pipeline (engine driver, dataset builder, model, confirmatory replication script, and all figure code) and configuration files are openly available under the MIT licence at https://github.com/tamalrkm/depth-of-satisficing. The pre-registrations are archived in the same OSF project (registrations 10.17605/OSF.IO/KB4ZQ and 10.17605/OSF.IO/A6NYK). Stockfish 18 and Maia-3 are third-party open-source software.

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**Competing interests.** The authors declare no competing interests.

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